General Concepts

General Terms

- **Bandwidth:** max data rate of a link (bits/sec).
- **Goodput:** proportion of successful slots (transmission without collision).
- **Throughput:** proportion of slots with a transmission (success or not).
- **Propagation time:** time for a signal to travel between two nodes.

Mathematical Foundations

Probability

- **Conditional:** $P(A \mid B) = \frac{P(A \cap B)}{P(B)}, P(B) > 0$
- **Bayes:** $P(A \mid B) = \frac{P(B|A)P(A)}{P(B)}$
- **Total Prob. Law:** $P(B) = \sum_i P(B \mid A_i)P(A_i)$ (partition $\{A_i\}$)
- **Expected Value:** $\mathbb{E}[X] = \sum_x xP(X = x)$
- **Memoryless:** $P(X > s + t \mid X > t) = P(X > s)$ (Geo/Exp).

Dist.	PMF	Notation	$\mathbb{E}[X]$	$\mathrm{Var}(X)$
Bernoulli	$P(X = 1) = p$	$\mathrm{Ber}(p)$	p	$p(1-p)$
Binomial	$\binom{n}{k} p^k (1-p)^{n-k}$	$\mathrm{Bin}(n, p)$	np	$np(1-p)$
Geometric	$(1-p)^{k-1}p$	$\mathrm{Geo}(p)$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Poisson	$\frac{\lambda^k e^{-\lambda}}{k!}$	$\mathrm{Pois}(\lambda)$	λ	λ

Computer networks vs Distributed systems

Circuit and Packet Switching

Timing

Multiplexing

The OSI Model (5-Layer model)

- **Physical:** Move bits on a link (signals, timing, encoding).
- **Link:** Frames on local network, MAC addressing, error detection.
- **Network:** Packet delivery across networks, routing (IP).
- **Transport:** End-to-end process communication, reliability/flow control (TCP/UDP).
- **Application:** End-user services/protocols (DNS, DHCP, HTTP, FTP etc.).

Link Layer & Local Area Networks (LAN)

Shared Medium Intuition

- Nodes broadcast on same channel; only one can transmit at a time.
- Simultaneous transmit $\Rightarrow$ collision $\Rightarrow$ data lost.
- Sender uses full link bandwidth while transmitting.

Definitions

- **Bandwidth:** max data rate of channel (B).
- **Throughput:** fraction of bandwidth used to send traffic.
- **Goodput:** fraction of bandwidth used for successful data.
- $\eta = \frac{\text{avg effective bandwidth}}{\text{total bandwidth}} = \frac{\text{avg effective time}}{\text{total time}}, 0 \leq \eta \leq 1.$

Multiple Access Domains

- **Multiple Access Domain:** set of nodes sharing a common channel.
- **Channel Partitioning:** divide channel into smaller "pieces" (time/freq/code).
- **Random Access:** transmit at will; handle collisions (ALOHA, CSMA).
- **Taking Turns:** nodes take turns to transmit (polling/token passing).

ALOHA protocol

- Random access on shared medium; collisions if overlap. Frame time T (size L, rate B): $T = L/B$.
- **Binomial model:** N users, each transmits in a slot w.p. p.
- **Poisson model:** aggregate attempts are Poisson with rate G (attempts/slot).
- **Slotted ALOHA (align to slots):**
  - Success if exactly one tx in a slot.
  - Binomial: $P(\text{succ}) = Np(1-p)^{N-1}$, max at $p = 1/N$ gives $S_{\max} \approx 1/e$.
  - Poisson: $S = Ge^{-G}$, $S_{\max} = 1/e$ at $G = 1$.
- **Pure ALOHA (no slots):**
  - Vulnerable period $2T$.
  - Binomial (approx): $P(\text{succ}) \approx Np(1-p)^{2(N-1)}$, max at $p \approx 1/(2N)$ gives $S_{\max} \approx 1/(2e)$.
  - Poisson: $S = Ge^{-2G}$, $S_{\max} = 1/(2e)$ at $G = 1/2$.

CSMA (Carrier Sense Multiple Access)

- Listen before tx; if channel idle, transmit entire frame; else wait.
- **Propagation delay** τ: time for signal to propagate between 2 farthest nodes.
- **CSMA/CD (Collision Detection):** abort tx on collision detection; reduces wasted time.
- **CSMA/CD efficiency:** $\eta = \frac{1}{1 + 5\frac{d_{\text{prop}}}{d_{\text{trans}}}}$.

CSMA/CD (Collision Detection)

- Detect collisions while transmitting; send jam; then backoff and retry.
- **Binary exponential backoff:** after m-th collision pick $k \in \{0, \ldots, 2^m - 1\}$, wait k slots.

CSMA/CA (Collision Avoidance)

- Used in Wi-Fi: cannot detect collisions reliably, so avoid them.
- If channel idle, wait DIFS then transmit; if busy, pick random backoff slots.
- Receiver replies with ACK after SIFS; missing ACK implies collision.

CSMA persistence

- **1-persistent:** transmit immediately when channel becomes idle (high collision under load).
- **0-persistent:** wait random backoff after busy (lower collision, higher delay).
- p**-persistent:** when idle, transmit with prob. p each slot (tradeoff).

Single (Logical) Link in practice

MAC Addresses

- Link-layer address unique per NIC (Network Interface Card); assigned by IEEE OUI.
- Used to deliver frames on same LAN; flat (non-hierarchical), portable across LANs.
- Usually burned-in, but can be software-set.

Ethernet

- Dominant wired LAN; many speed generations; cheap NICs.
- **CSMA/CD (classic shared medium):**
  1. Sense channel; if idle transmit, else wait.
  2. If collision detected while tx, abort and send jam.
  3. **Binary exponential backoff:** after m-th collision pick $k \in \{0, \ldots, 2^m - 1\}$, wait $k \cdot 512$ bit-times, retry.

Interconnecting broadcast domains

- **Switch:**
  - link-layer device that breaks broadcast domains; stores/forwards frames using MAC table.
  - Transparent to hosts; self-learning (no manual config), entries age out (TTL).
  - Each link is its own collision domain; enables simultaneous transmissions.
  - Forwarding: if dest known on incoming port $\Rightarrow$ drop; else forward to port; if unknown $\Rightarrow$ flood.
  - **Switch table:** (MAC, port, TTL). Learn from source MAC+ingress port on each frame.
- **Loops:** cause broadcast storms, multiple frame copies, MAC table instability.
- **Spanning Tree Protocol (STP):** disable links to eliminate loops; elect root, compute shortest-path tree.
- **HELLO/BPDU fields:** SID (switch ID), RID (root ID), DTR (distance to root).
- Root selection: prefer lower RID; then lower DTR; then lower SID (tie-break).

"Layer 2.5" - ARP

ARP is generally considered to be part of Layer 2 (Data Link Layer), but it works directly to support Layer 3 (Network Layer).

- Why it's Layer 2: ARP messages are encapsulated directly into Ethernet frames. They do not have an IP header (Layer 3) and they cannot be routed outside of your local network (LAN).
- Why it feels like Layer 3: Its entire purpose is to handle Layer 3 information (IP addresses). It is the mechanism that allows Layer 3 to actually function on physical hardware.

Address Resolution Protocol (ARP)

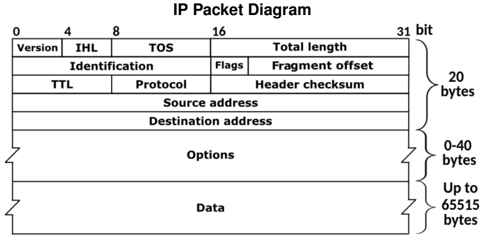
- Translates between MAC addresses and IP addresses on a LAN.
- Needed because Layer 3 provides the IP, but frames on Layer 2 need a destination MAC.
- To send outside the LAN, the packet uses the IP of the destination but the MAC of the next-hop router; routers then forward using their own L2 on each hop.
- Hosts use DHCP and routing to know whether the destination IP is on the same LAN or remote.
- Each node maintains an **ARP cache** (IP $\rightarrow$ MAC table).
- If the mapping is missing, send an **ARP request** (broadcast in the local broadcast domain).
- The target replies with an **ARP response** containing its MAC, which is stored in the cache.
- ARP messages stay within a single broadcast domain (not routed).

Network Layer (IP & Routing)

Introduction to IP addressing

- IPv4 address = 32 bits (a.b.c.d), identifies interface (host/router can have multiple interfaces).
- Subnet vs host split via CIDR: a.b.c.d/x where x = prefix length (subnet mask).
- Example: 192.168.1.178/24 $\Rightarrow$ subnet 192.168.1, host 178.
- Hierarchical addressing enables routing scalability; ISPs advertise prefixes.
- Forwarding uses **longest prefix match**.
- **Address discovery protocols:**

IP Packet Diagram



Transport Layer (TCP & UDP)

Application Layer (HTTP, DNS, SMTP & FTP)

DNS (Domain Name System)

Maps between hostnames and IP addresses. Each server is authoritative for its domain and can resolve queries or forward them; resolution ends when an authoritative server replies or the requested data is found in cache.

Record types:

- **A**: maps an Internet hostname to an IP address.
- **NS**: maps a domain name (e.g., `google.com`) to the server responsible for it.
- **CNAME**: maps a hostname to its canonical name, e.g., `wedsac.mit.com` is actually `www.mit.com`.
- **PTR**: reverse of A (IP address to hostname).
- **MX**: maps a hostname to the mail server that handles it.

DNS queries and replies use UDP and typically port 53 (smaller packets for requests/replies).

Servers return a **TTL**; the answer is cached for that duration.

Glued response: if the client needs to query the authoritative server but does not know its IP, the server returns the IP in the NS response.

Security issues:

1. **DDoS Attack**: flooding DNS servers so they cannot answer other requests.
2. **MITM (Man-in-the-Middle) Attack**: when a client queries a server, a fake response can arrive before the real one, potentially mapping a requested name to a desired IP.
3. **Cache poisoning**: if the server is not well-protected, it continues serving an injected response until TTL expires; thus, we must ensure the reply is authentic.
4. **Kaminsky attack**: attacker sends many requests for random subdomains of a target domain (e.g., `random1.target.com`, `random2.target.com`, etc.) to a resolver; when the resolver queries the authoritative server, the attacker floods it with fake replies, hoping one matches the query ID and port, poisoning the cache.

- **DNSSEC**: adds signatures so you can verify who answered and add more security; adds extra information and expands UDP packet size.

DHCP (Dynamic Host Configuration Protocol)

- Allows host to obtain IP dynamically (no manual config).
- Each time host connects, it can get a different IP; server can **lease** addresses and reuse later.
- Used by devices joining a network.
- **Message flow (DORA):**
  - **DHCP Discover:** src IP 0.0.0.0, dst 255.255.255.255 (broadcast, server unknown).
  - **DHCP Offer:** server replies, dst 255.255.255.255 (broadcast); includes **yiaddr** (your IP).
  - **DHCP Request:** client broadcasts 0.0.0.0 $\rightarrow$ 255.255.255.255 to accept one offer; includes chosen **yiaddr** so other servers withdraw.
  - **DHCP ACK:** server confirms lease; may broadcast so others know address in use.
- **DHCP relaying:** if DHCP server not on local subnet, relay forwards messages to server on another network.

Interdomain Routing & BGP

Autonomous Systems (AS)

- **Definition:** IP networks/routers under one admin with a common routing policy.
- **ASN:** Unique 16/32-bit AS identifier used in BGP.
- **AS relationships:**
  - **Customer** / **Provider:** Customer pays provider for transit.
  - **Full** / **Partial transit:** Full = global reachability; partial = limited reach.
  - **Peering:** Settlement-free; advertise own + customers only.
- **AS-level topology:** Common categories
  - **Tier-1:** No providers; global reach via peering.
  - **Tier-2:** Providers + peers (buys some transit).
  - **Stub AS:** Single provider; no downstream customers.

Border Gateway Protocol (BGP)

- **Definition:** Inter-AS routing protocol.
- **Path-vector:** Advertises AS_PATH; reject if own ASN appears.
- **Policy + attributes:** LOCAL_PREF, MED, AS_PATH, NEXT_HOP, etc.

BGP message flow (5 stages)

1. **TCP setup:** Session on TCP/179.
2. **OPEN:** Exchange ASN, hold-time, capabilities.
3. **Initial UPDATE:** Send NLRI + attributes.
4. **Decision:** Import policy + best-path selection.
5. **Steady state:** KEEPALIVE + incremental UPDATE/NOTIFICATION.

iBGP, eBGP and IGP interaction

- **eBGP:** Between ASes; increments AS_PATH, may change NEXT_HOP.
- **iBGP:** Within AS; no AS_PATH prepend; needs full mesh or route-reflectors.
- **IGP:** Provides internal reachability to BGP NEXT_HOP.

BGP Stability

- **Stable state:** Routes converge with no further changes.
- **Safety theorem (informal):** Under Gao-Rexford, BGP converges to a stable outcome.
- **Gao-Rexford conditions (commercial routing)**
  - **Topology:** Customer-provider edges form a DAG; peers are lateral.
  - **Preference:** Customer > peer > provider.
  - **Export:** Export all routes to customers; only customer routes to peers/providers.